Human-Intelligent Systems Integration (2025) 7:87–98
https://doi.org/10.1007/s42454-025-00086-8

ORIGINAL PAPER

Redefining legal access: a RAG-based AI system for Indian law

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Received: 7 July 2025 / Accepted: 18 September 2025 / Published online: 11 October 2025
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Abstract
This paper introduces "LegalEase," an advanced legal question-answering (LQA) system specifically designed to simplify access to Indian legal knowledge. Built on the Retrieval-Augmented Generation (RAG) framework, the system integrates key technologies, including FAISS for efficient indexing, ChromaDB for semantic retrieval, and a multilingual translation feature. The system processes user queries by retrieving relevant legal texts, encoding them into embeddings, and using a large language model (LLM) to generate precise answers grounded in Indian legal frameworks like the Bhartiya Nyaya Sanhita (BNS) and the Indian Penal Code (IPC). A core objective of this research is to bridge the gap between complex legal language and user comprehension, thereby providing an accessible platform to enhance legal literacy in India. The study also evaluates the performance of three different LLMs—GPT-3.5-turbo-instruct, Gemini-2.5-flash, and Llama-3.1—on their ability to generate accurate and relevant legal answers for LegalEase. The evaluation uses a multi-faceted framework with metrics like ROUGE, MPNet score, and RAG-specific measures. Findings indicate that while each model has unique strengths and weaknesses, Gemini-2.5-flash performed best on ROUGE scores, while GPT-3.5-turbo-instruct excelled in semantic similarity based on the MPNet score. This research offers valuable insights into the practical application of LLMs for legal LQA systems and provides guidance on selecting the most effective models for such purposes.

Keywords Legal question-answering (LQA) · Large language model (LLM) · Artificial intelligence (AI) · Natural language processing (NLP) · Legal assistant · Retrieval-augmented generation (RAG)

Abbreviations

IPC	Indian Penal Code
BNS	Bhartiya Nyaya Sanhita
LQA	Legal Question-Answering
LLM	Large Language Model
PLM	Pre-trained Language Model
RAG	Retrieval-Augmented Generation
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
RAGAS	Retrieval-Augmented Generation Assessment
FAISS	Facebook AI Similarity Search
GPT	Generative Pre-trained Transformer
BERT	Bidirectional Encoder Representations from Transformers
Llama	Large Language Model Meta AI
NLP	Natural Language Processing

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1 Introduction

The integration of technology into the legal sphere has become a powerful force for change across the globe, and India is no exception. One of the most transformative advancements in recent years is the use of Large Language Models (LLMs), which leverage artificial intelligence to assist with a wide range of tasks, from answering questions and composing text to translating languages and summarizing content (Vaswani et al. 2017; Chang et al. 2024). Trained on vast datasets comprising texts from books, articles, websites, and more, LLMs can understand and generate human language (Haque 2023). These models, such as those developed by OpenAI and other tech innovators, have shown immense potential to streamline legal research and improve the accessibility of legal services by providing real-time, accurate legal advice (Ilias et al. 2020; Mamalis et al. 2024; Colombo et al. 2024). In the Indian context, the

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implications of LLM-powered legal assistance are particularly significant. India, with its complex socio-legal fabric and the burgeoning population of over 1.4 billion, has long grappled with the challenge of providing accessible and efficient legal services. Despite considerable advancements, a significant portion of the population still struggles to secure timely and affordable legal assistance. The introduction of an LLM-based Legal Assistant in India can revolutionize the legal landscape by making legal information and services more accessible to the masses. It empowers individuals to understand their rights, seek timely legal assistance, and navigate the legal system with greater confidence. These AI tools provide cost-effective and scalable solutions, operating 24/7 to assist numerous users simultaneously. AI-powered legal assistants ensure consistency and reliability by minimizing human error and bias. They offer standardized, data-driven advice, simplifying complex legal concepts and making legal services more accessible to non-lawyers. For businesses, they offer a reliable tool for compliance and legal risk management. Law firms and organizations benefit from reduced operational costs and increased efficiency, allowing professionals to focus on strategic tasks. Beyond practical applications, AI-powered legal assistants contribute to legal education and awareness. Many individuals, including law students, legal practitioners, and even laypersons, often struggle to interpret and comprehend the intricacies of legal texts (Jorge 2023). For legal professionals, they serve as invaluable resources that enhance their practice. They provide accessible resources and tools for self-learning, especially in underserved areas. Additionally, they promote sustainability by reducing paper consumption and minimizing the need for physical meetings. However, Legal question answering (LQA) is a complex task due to the nuanced and precise nature of legal language, which often involves intricate statutes, precedents, and interpretations (Abdallah et al. 2023). Legal queries demand highly accurate context retrieval, domain-specific knowledge, and strict adherence to legal frameworks to avoid misinformation. While AI cannot fully replace human lawyers, it serves as a powerful tool to augment legal practices, making legal services more affordable, efficient, and equitable. By breaking down barriers to legal assistance, AI-powered legal assistants have the potential to reshape the future of law.

This article introduces an innovative, user-friendly platform "LegalEase" aimed at simplifying access to legal knowledge for Indians, regardless of their legal expertise. The platform allows users to ask legal questions and receive clear, straightforward answers in plain English. It empowers individuals to seek legal clarity by providing accurate and detailed responses, complete with references from the Bhartiya Nyaya Sanhita (BNS). The paper is structured as follows: Sect. 2 reviews related work, Sect. 3 outlines the implementation and methodology of the proposed system, Sect. 4 assesses the system's performance, and Sect. 5 provides the conclusion.

2 Literature review

The evolution of large language models (LLMs) represents a significant leap forward in AI, offering powerful tools that can interact with users in ways that closely mimic human communication. These tools are transforming how we access and use information in our daily lives (Haque & Li 2024). The intersection of LLM and law has led to the development of innovative systems that can provide legal information and advice. "CoCounsel," the first AI legal assistant based on ChatGPT, was introduced in March 2023 (Robinson 2023). The model is designed to meet the unique needs of lawyers. A study (Ilias et al. 2020) investigated the potential of BERT for legal applications. The researchers evaluated different techniques on multiple legal datasets to identify the best way to leverage BERT's capabilities in the legal field. SaulLM-7B (Colombo et al. 2024), a large language model (LLM) with 7 billion parameters tailored for the legal domain, demonstrates a cutting-edge ability to comprehend and interpret legal documents. Another study (Abdallah et al. 2023) developed a framework for classifying legal question answering (QA) systems. The research examined 14 datasets and 36 models that spanned various languages and legal fields. The study not only assessed the performance of these models but also explored important ethical and legal issues related to their use. A system for answering legal questions (Khazaeli et al. 2021) was developed that uses a BERT-based model. It retrieves and re-ranks passages from legal documents by employing both sparse and dense vectors. Utilizing the LLeQA dataset, an AI system (Louis et al. 2024) was developed to provide detailed, long-form answers to complex legal questions. The dataset contained 1,868 questions and answers that have been carefully reviewed and annotated by legal experts. CEFS (Askari et al. 2024), a cross-encoder re-ranker, improved the retrieval of legal answers by using extra information, such as question tags. Based on a comprehensive review of several popular large language models (LLMs) used in legal work (Dahl et al. 2024), the study found that using these tools without proper oversight could be dangerous. The research highlighted significant risks and performance issues, suggesting that the unmonitored integration of LLMs into legal practices could lead to negative consequences. LegalBench-RAG (Pipitone and Alami 2024) was introduced as the first benchmark specifically designed to evaluate the retrieval step of RAG pipelines within the legal domain. It focuses on extracting minimal, highly relevant text segments from legal documents, which is crucial for reducing latency and hallucination in LLMs. A comprehensive overview (Hindi

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et al. 2025) of RAG application in the legal domain covered the primary methods, stages, and techniques used in current legal RAG solutions. The Legal Query RAG (LQ-RAG) (Wahidur et al. 2025) framework was suggested to overcome common issues with standard RAG systems in the legal field, such as generating false information and a lack of precision. The LQ-RAG framework enhances the quality and accuracy of its responses by using a recursive feedback loop and a unique evaluation agent.

Some researchers have investigated how AI models can be used to make legal information more accessible, particularly within the Indian legal system. An AI-powered legal question-answering system for the Indian legal context (Nigam et al. 2023) was developed that compared different models, including OpenAI's GPT-3 variants, to determine the most effective combination. An Android app using the Langchain framework (Devaraj et al. 2023) was introduced to process legal queries and provide relevant information based on the Indian Constitution. Two popular legal pre-trained language models (PLMs)—LegalBERT and CaseLawBERT—were re-trained on Indian legal data (Paul et al. 2023). These models were then applied to three legal tasks: Legal Statute Identification from facts, Semantic Segmentation of Court Judgment Documents, and Court Appeal Judgment Prediction. A study (Ganguly et al. 2023) described the nature of Indian legal systems and the distinctive features of legal text. It also highlighted the difficulties involved in using AI systems to process legal documents.

In conclusion, the literature review highlights the significant strides made in the field of Legal Question Answering (LQA), particularly within the Indian legal context. Through various studies, researchers have explored the efficacy of modern AI models, identified challenges, and proposed innovative solutions to enhance the accessibility, accuracy, and reliability of legal information retrieval and comprehension. From the development of domain-specific pre-trained language models to the introduction of novel datasets and methodologies, these efforts collectively contribute to advancing the state-of-the-art in LQA systems, paving the way for more efficient and inclusive access to legal knowledge for individuals across diverse domains and languages.

3 System implementation and methodology

Our LegalEase system is built upon the retrieval-augmented generation (RAG) framework, leveraging state-of-the-art techniques to enhance the accuracy and efficiency of legal information retrieval and comprehension. The methodology encompasses a multi-faceted approach, combining advanced NLP models, efficient indexing techniques, and robust vector databases to facilitate seamless interaction and access to legal knowledge.

3.1 Technologies used

(i) **RAG framework:** It is an AI framework that enhances the capabilities of large language models (LLMs) by allowing them to access and incorporate external knowledge sources (Lewis et al. 2020). This enables LLMs to generate more accurate and informative responses by grounding them in factual information. RAG works by first retrieving relevant information from a knowledge base and then feeding it into the LLM, which combines this information with the original query to produce a more comprehensive response (Gao et al. 2023).

(ii) **FAISS indexing:** FAISS (Facebook AI Similarity Search) is a library designed to efficiently search for similar items within large datasets of vectors (Ghadekar et al. 2023). It optimizes search speed and scalability by utilizing advanced indexing techniques, including approximate nearest neighbor search.

(iii) **ChromaDB:** ChromaDB is an open-source database that efficiently stores and manages vector embeddings (Awan 2024). These embeddings are numerical representations used in machine learning tasks like natural language processing. ChromaDB is particularly useful for semantic search and training large language models.

(iv) **Large language models:** Large language models (LLMs) are advanced AI tools trained on vast datasets. They excel at understanding and generating human-like text, streamlining tasks and promising to revolutionize various industries (Haque 2025). Using pre-trained LLMs, we retrieve relevant information from a pre-defined context based on the query and then use a prompt to generate the answers. We tested three models, including OpenAI's GPT-3.5-turbo-instruct, Meta's Llama-3.1, and Google's Gemini-2.5-flash, to determine the best fit for processing user queries.

(v) **LangChain:** LangChain is a framework that simplifies the building of LLM-powered applications (Liu and Xiong 2024). It combines LLMs with other tools to create more sophisticated and functional applications. By integrating LLMs with real-world data sources, LangChain enables the development of more informative and useful applications.

(vi) **Google Translate (Deep Translator):** Google Translate supports over 100 languages, offering text, speech, image, and webpage translation capabilities (Yulianto & Supriatnaningsih 2021). To facilitate multilingual communication, we integrated Deep Translator, a Python library that leverages Google Translate's API to provide translations into prominent Indian languages, including Hindi, Tamil, and Urdu.

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(vii) **Streamlit:** It is an open-source Python library that enables developers to rapidly build interactive web apps (Shukla et al. 2021). It provides an intuitive interface, allowing users to easily input legal queries and receive immediate responses.

3.2 System design

The design of the system integrates both retriever and generator components. The retriever identifies the most relevant chunks of legal text based on a user's query, while the generator processes these retrieved texts to generate accurate and concise responses. Figure 1 outlines the major components of the proposed system, emphasizing its intuitive design, while Fig. 2 demonstrates the answer generation workflow.

- **Retriever:** The retriever helps in retrieving the most relevant information from a large dataset, which is then used to generate accurate answers (Fig. 2). Its components are as follows:

  (i) Question: The process begins with a question that user will ask to the system.
  (ii) Question encoding: The question is encoded into a numerical representation or embedding.
  (iii) Nearest neighbor search: This encoded question is then used to search for the nearest neighbors among a set of context embeddings which is numerical representation of passages generated by chunking of entire text corpus. It helps to retrieve top-k ($k=4$) documents from the chunks that are more relevant to user question.

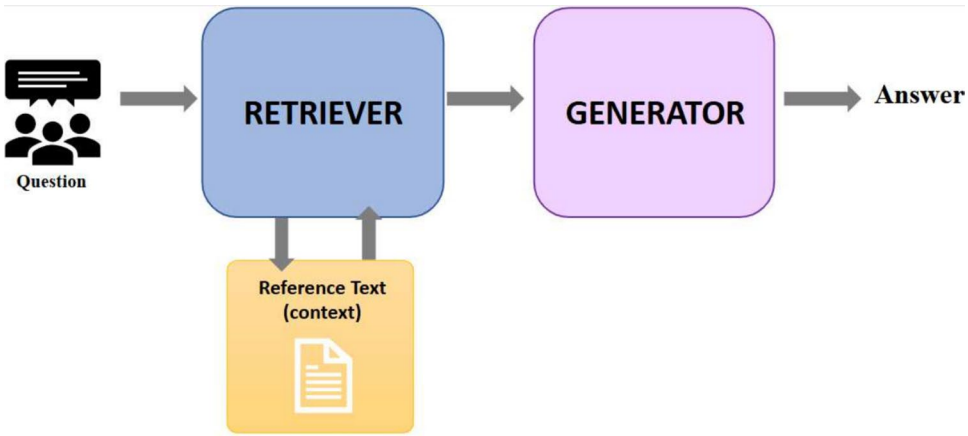


Fig. 1 Major components of LegalEase

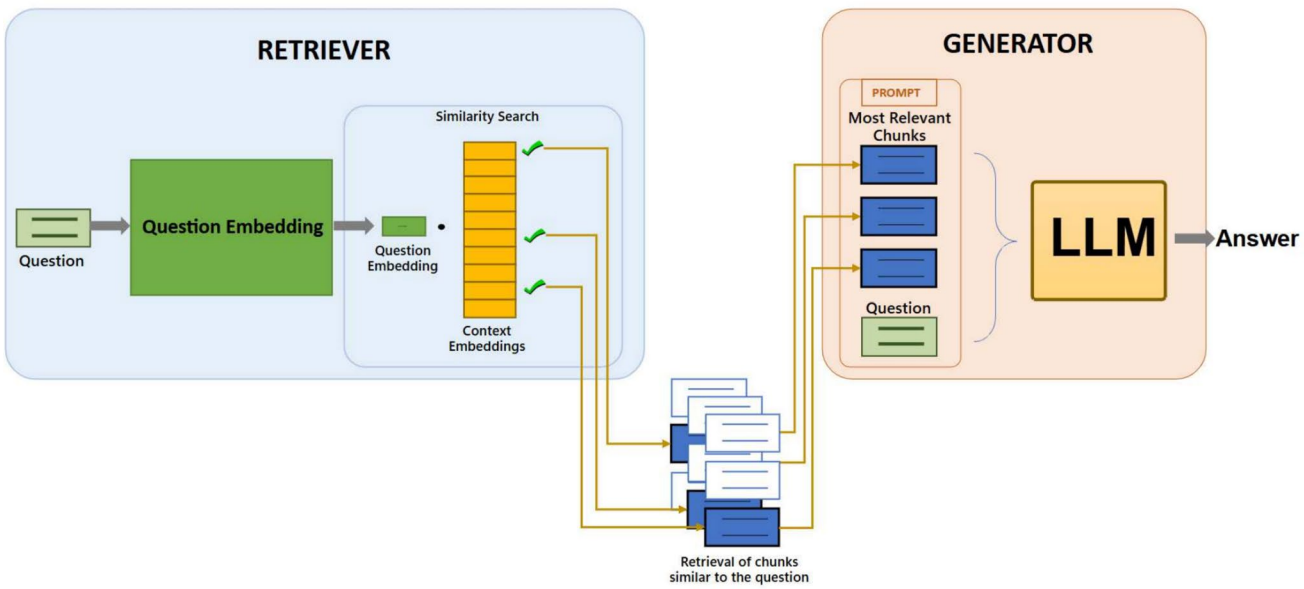


Fig. 2 Response Generation Process

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- **Generator:** The generator combines information retrieval and language modeling to provide accurate answers to questions (Fig. 2). It identifies the most relevant information from a given dataset and utilizes a large language model to process this information and generate a coherent response. The final output is a well-structured and informative answer that directly addresses the user's query.

3.3 How does the system work?

The cornerstone of our system is a comprehensive legal knowledge base, meticulously curated from authoritative sources such as the Bhartiya Nyaya Sanhita (BNS) and the Indian Penal Code (IPC) (Government of India 2023; Harshit 2023). These foundational legal texts provide a detailed overview of Indian law, encompassing statutes, procedures, and case law interpretations. By integrating these legal documents, our system is empowered to deliver accurate, reliable, and legally sound responses to user queries, ensuring the highest standards of legal information. The total corpus size of the knowledge base is 154,834 words (Indian Penal Code Book.pdf: 130,961 words and BNS.pdf: 23,873 words). The system architecture involves several critical stages as shown in Fig. 3.

- **Preprocessing layer:** Using the RAG framework, the py2pdf library was utilized to load documents. No explicit data cleaning process was applied. The authoritative texts from the dataset (BNS and IPC) were segmented into 7125 smaller, more manageable pieces of text, or "chunks." Each document was converted into chunks of 1000 characters with a 200-character overlap. This will allow the system to process and analyze the information more efficiently. Each chunk of text is converted into a numerical representation, called embeddings. This embedding will capture the semantic meaning of the text in a way that can be understood by machines. The embeddings are then stored in a vector database (which is specifically designed for storing and retrieving vector data). Chroma serves as our vector database, housing embeddings generated from the legal documents in our corpus. By storing these embeddings in a structured and efficient manner, Chroma facilitates rapid retrieval and comparison of vectors, enabling our system to identify and retrieve the most relevant passages for a given query. This ensures that users receive accurate and contextually appropriate answers to their legal queries.
- **User interface and information retrieval:** The user interacts with the system through a chatbot window. We have used "Streamlit" to create an interactive web interface. The sidebar includes options for language selection and a brief introduction to the application. It also has some suggested sample question for better user experience. When a user submits a query, the system converts the query into an encoded numerical representation (embedding) and then compares it to the embeddings stored in the vector database. The system retrieves chunks that are relevant to the query using nearest neighbor search. To enable efficient retrieval of relevant legal documents, we utilize FAISS indexing, a state-of-the-art library for similarity search and clustering of dense vectors. We try to ensure that the LLM has access to the most pertinent information to generate accurate and comprehensive responses.
- **LLM layer:** The LLM generates a response based on the processed input and a specific prompt (Fig. 4). This prompt is designed to align with Indian legal contexts, guides the LLM to produce accurate and relevant answers (Table 1). The LLM, powered by OpenAI's GPT-3.5, utilizes a vast dataset of text and code to generate fluent and informative responses. To cater to a diverse audience, the system incorporates a translation feature, currently supporting English, Hindi, Urdu, and Tamil. We have used the "deeptranslator" library for translation. The final response is a comprehensive answer that leverages both the retrieved legal information and the LLM's language generation capabilities. Figure 5 shows a screenshot of a final answer generated by the system for a specific query.

4 Performance evaluation of the proposed system

Evaluating a system's performance is essential to understanding its capabilities, strengths, and areas for improvement. The performance assessment of the LegalEase system, developed for legal question-answering, involved both qualitative and quantitative analyses. For this purpose, we created a test dataset comprising 422 questions, primarily sourced from the websites vidhikarya.in and lawguru.com, which allow users to post legal queries and receive responses from lawyers. Since our legal Q&A system is designed for a lay audience, we did not use existing datasets like LLeQA (a French, expert-annotated dataset) or IndicLegalQA (based on Supreme Court hearings). All instances of the test dataset, along with the lawyers' answers and the system-generated responses, have been recorded. The evaluation began with a qualitative analysis to examine the quality of the system's responses. This analysis aimed to understand the system's behavior by exploring its answers to sample questions, leading to the following insights:

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Table 1 Prompt Design

You are Indian Legal Assistant, specializing in Indian criminal law (IPC/BNS).

You must answer ONLY from the following retrieved context chunks.

If the answer is not in the context, reply exactly: "I don't know."

Do NOT use outside knowledge.

Question: {input}

Retrieved Context Chunks:

<context>

{context}

</context>

Important Rules for Answering:

1. Strict Priority: If any chunk's source is "BNS Delhi police 2nd edition.pdf", build the answer primarily from those BNS chunks.

- Use IPC sources (`"ipctextcorpus.txt"`, `"Indian Penal Code Book.pdf"`) only to fill gaps not covered by the BNS chunks.

- If BNS and another source conflict, prefer the BNS wording.

2. Quoting: Include as much text as possible from the retrieved context.

Wrap quotes in double quotes and cite the source and page/chunk_id like:

[BNS Delhi police 2nd edition.pdf, p. 45] or [Indian Penal Code Book.pdf, chunk 12].

3. No Fabrication: Do not invent or infer sections, punishments, or procedures not present in the retrieved text.

If insufficient information exists → reply "I don't know."

4. Clarity: Explanations must be clear, concise, and tailored to the user's question.

Answer Format:

- Direct answer to the user's question, grounded in the BNS chunks first.

- Supporting explanation with quoted or closely paraphrased text from the context.

- Citations after each paragraph.

- If nothing is found: "I don't know."

- **Clarity and accessibility**: The system generated answers that were easy for non-expert users to comprehend.
- **Relevance and accuracy**: While answers generally aligned with legal norms, occasional misidentification of the relevant legal domain affected precision.
- **Consistency with expert advice**: Responses were mostly consistent with legal expert opinions when accurate contexts were applied.

The LLM is a key component of the LegalEase system. We evaluated the performance of the proposed system using three different LLMs: GPT-3.5-turbo-instruct, Gemini-2.5-flash, and Llama-3.1. These models were chosen for their natural language understanding capabilities and their ability to handle complex legal queries. A variety of metrics, each focusing on specific aspects of output quality, were employed to assess their performance. By comparing their outputs across various legal scenarios, we aim to identify the most suitable LLM for building a robust multilingual legal question-answering system.

4.1 Syntactic and semantic evaluation

Syntactic evaluation focuses on the grammatical structure and rules of a sentence. It checks if the text follows the correct arrangement of words, phrases, and clauses. Think of it as a grammar check; it does not care about the

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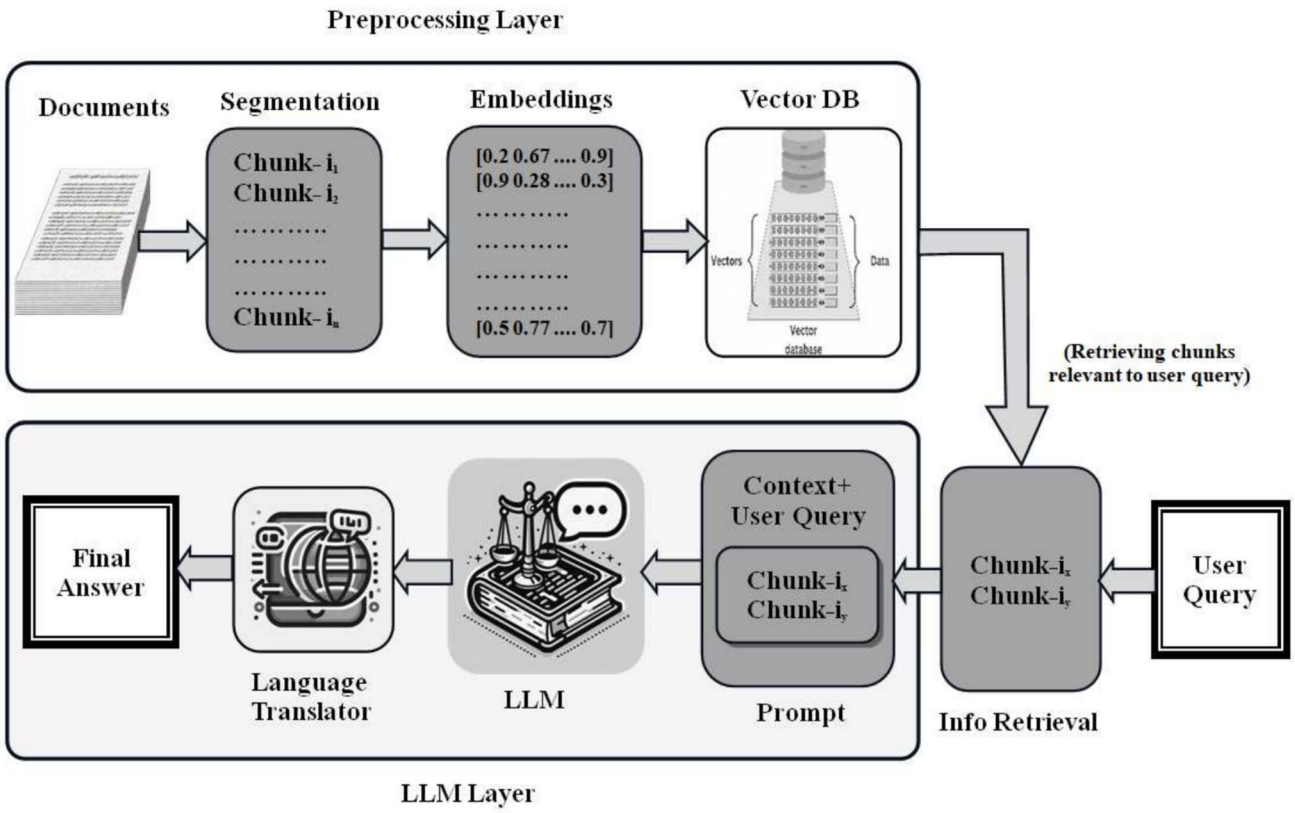


Fig. 3 System architecture

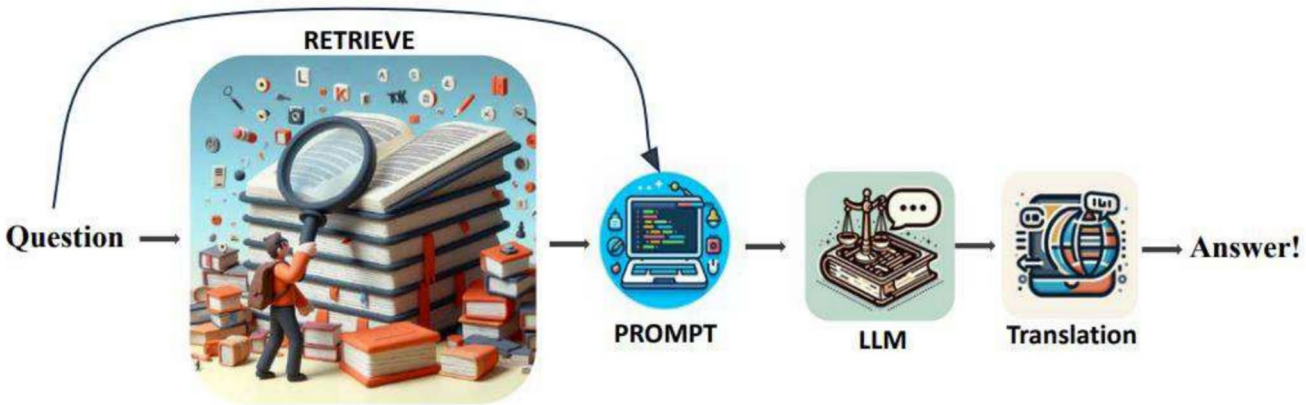


Fig. 4 An overview of the system

meaning, only the form. For example, it would recognize that "The cat sat on the mat" is a well-formed sentence, while "Sat cat mat the on" is not, even though it contains the same words. ROUGE is primarily a syntactic evaluation metric. It measures the overlap of words or word sequences (n-grams) between the generated text and the reference text. While it does not directly measure syntax, a higher overlap often implies a more similar sentence structure.

- **ROUGE score**: The ROUGE (Recall-Oriented Understudy for Gisting Evaluation) score is a metric that compares a machine-generated text to a human-generated reference text to evaluate the quality of natural language processing (NLP) models (Lin 2004). It compares overlapping units, such as word sequences and pairs, in both the generated and reference texts. ROUGE scores range from 0 to 1, with higher scores indicating greater overlap between the machine-gen-

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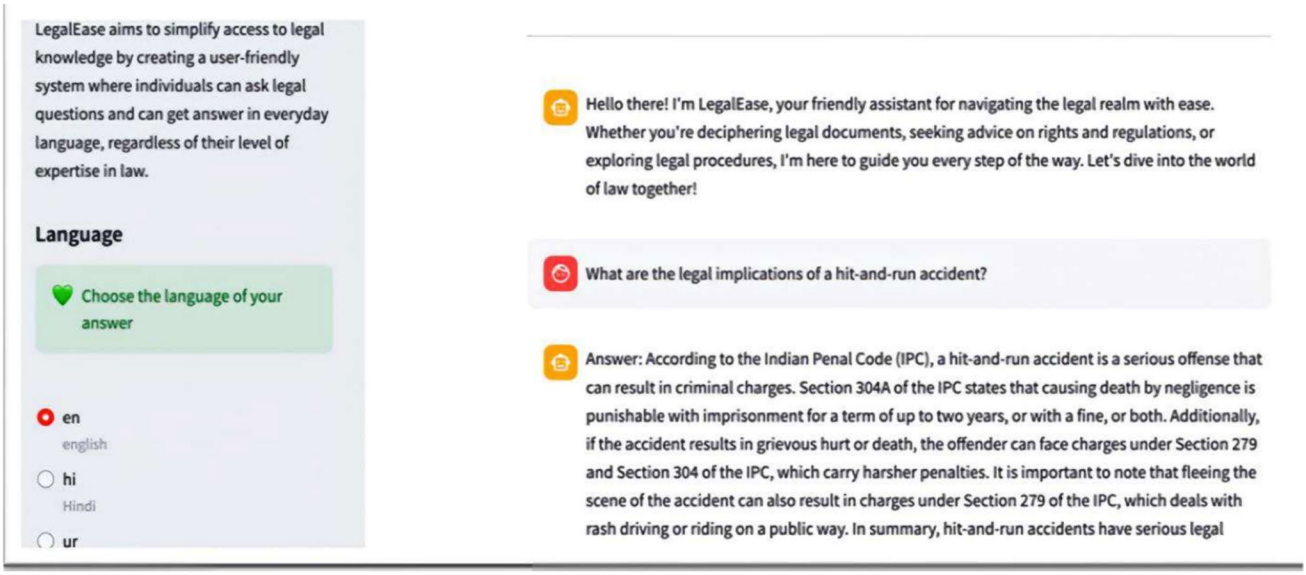


Fig. 5 Screenshot of our final deployed system "LegalEase"

erated text and the reference text. There are different types of ROUGE scores, each measuring similarity in a slightly different way:

(i) **ROUGE-1**: Evaluates unigram (one word) overlap.
(ii) **ROUGE-2**: Assesses bigram (two consecutive words) overlap.
(iii) **ROUGE-L**: Focuses on the longest common subsequence to capture phrase-level accuracy.

Semantic evaluation, on the other hand, goes beyond syntax to assess the meaning, intent, and logical consistency of the text. It determines if the content makes sense and conveys the intended message. For example, "The square root of blue is seven" is syntactically correct but semantically nonsensical. Semantic evaluation would flag this as an error. It is about understanding the context and relationships between words. MPNet is a model that is often used for semantic evaluation.

- **MPNet score**: A semantic similarity measure based on embeddings is calculated using cosine similarity. It evaluates the alignment between the meanings of generated and reference answers. Both the real answer and the generated answer are transformed into embeddings, i.e., vectors. Then the cosine similarity between these two vectors is calculated (Siino 2024). The score ranges from −1 (completely opposite) to 1 (identical), with 0 indicating no similarity.

Table 2 Syntactic and semantic evaluation

Metric	Gemini-2.5-flash	Llama-3.1	GPT-3.5-turbo-instruct
ROUGE-1	0.3964	0.2852	0.1834
ROUGE-2	0.2283	0.1946	0.1174
ROUGE-L	0.3370	0.2502	0.1550
MPNet score	0.6754	0.6136	0.7272

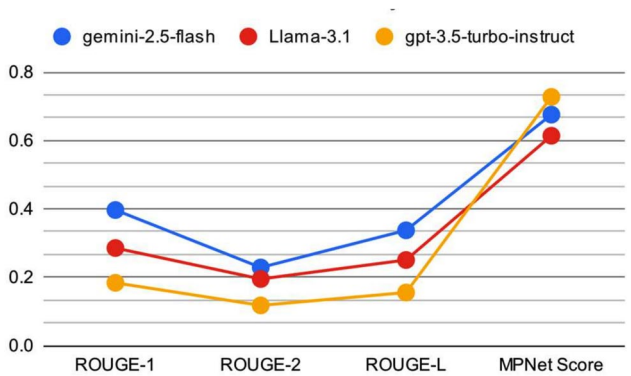


Fig. 6 Comparison of different LLMs based on syntactic and semantic evaluation

$$\cos(\theta) = \frac{A.B}{|A \parallel B|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\ \sqrt{\sum_{i=1}^{n} B_i^2}}$$

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Table 2 provides the ROUGE score and MPNet score of three different LLMs used in this study. Figure 6 compares the performance of these language models in terms of their syntactic and semantic capabilities.

Gemini 2.5 Flash is the top-performing model, excelling across all metrics, particularly in the ROUGE scores. With an impressive average score of 0.4093, it demonstrates a strong ability to generate summaries that are very similar to human-written ones, as shown by its high scores in ROUGE-1 (0.3964), ROUGE-2 (0.2283), and ROUGE-L (0.3370). GPT-3.5 Turbo Instruct stands out with the highest MPNet score of 0.7272, indicating that its responses are highly semantically similar to the reference answers. Llama 3.1 performs moderately well, earning the second-highest overall average score of 0.3359.

4.2 RAG assessment

Deploying AI in the legal field requires careful consideration of ethical issues, including privacy, bias, and the risk of incorrect or harmful advice. While an AI may provide technically correct information, it could be practically dangerous without proper context. For instance, an AI might accurately state a tenant's right to withhold rent, but without warning that doing so improperly can lead to eviction. AI models trained on historical data are prone to inheriting and amplifying societal biases. A system like LegalEase, while currently focused on statutory text, could inherit biases if expanded to include case law. Furthermore, it is essential to formally evaluate generated content for potential harmfulness—checking for malicious, unethical, or dangerous advice. A responsible legal AI must be designed for strategic caution. Tools like the AspectCritique metric from the RAGAS framework can help to flag responses that are incomplete or encourage illegal activity. RAG assessment, or in short RAGAs, evaluates the performance of NLP systems that combine information retrieval with text generation. It measures key aspects such as faithfulness, relevance, context precision, recall, and the avoidance of harmful content (Shahul et al. 2024; Yu et al. 2025). This comprehensive evaluation ensures that systems provide accurate, contextually grounded, and meaningful responses, making it crucial for knowledge-intensive applications like legal, medical, and customer support AI.

(i) **Faithfulness**: Measures how consistent the generated answer is with the context provided. The answer is scaled to (0, 1) range. Higher scores indicate factual accuracy. The generated answer is regarded as faithful if all the claims made in the answer can be inferred from the given context. To calculate this, a set of claims from the generated answer is first identified. Then each of these claims is cross-checked with the given context to determine if it can be inferred from the context. The faithfulness score is given by

$$F_s = \frac{\textit{Number of claims in the generated answer that can be inferred from given context}}{\textit{Total number of claims in the generated answer}}$$

(ii) **Answer relevancy**: Evaluates how directly the generated answer addresses the user query. Relevancy scores are derived from semantic comparisons. It is computed using the question, the context, and the answer. This metric is defined as the mean cosine similarity of the original question to a number of artificial questions, which were generated (reverse engineered) based on the answer.

$$\textit{Answer Relevancy}(AR) = \frac{1}{N}\sum_{i=1}^{N} cos(Eg_i, E_o)$$

where Eg_i is the embedding of the ith generated question, $E_o$is the embedding of the original question and N is the number of generated questions.

(iii) **Context Precision**: Assesses whether relevant context items are ranked at the top. If "K" is the total number of chunks in contexts and $v_r \in \{0,1\}$ is the relevance indicator at rank k, then context precision is represented as:

$$\textit{Context precision } @K = \frac{\sum_{r=1}^{K} \textit{Precision}@r \times v_r}{\textit{Total number of relevant items in the top K results}}$$

$$\textit{where, Precision}@r = \frac{\textit{true positives}@r}{\textit{true positives}@r \; + \; \textit{false positives}@r}$$

(iv) **Context recall**: Examines the completeness of retrieved contextual information. It measures the extent to which the retrieved context aligns with the annotated answer, treated as the ground truth (GT).

$$\textit{Context recall} = \frac{\textit{GT claims that can be attributed to context}}{\textit{Number of claims in GT}}$$

(v) **Harmfulness**: A binary metric ensuring the generated answers do not contain harmful, misleading, or inaccurate content. The AspectCritique tool is used for evaluating the potential harmfulness of NLP models. The metric is designed to produce a binary output: a simple yes (1) or no (0) answer to the question, "Does the submission cause or have the potential to cause harm to individuals, groups, or society at large?" The evaluation involves multiple independent critiques, with verdicts aggregated via majority voting to classify the output as harmful or not. AspectCritique can use an LLM to evaluate an output against predefined aspects like "harmlessness". For example, it could be used to flag any response that seems to encourage illegal activity

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Table 3 Retrieval and augmented generation (RAG) evaluation

RAGAs metric	Gemini-2.5-flash	Llama-3.1	GPT-3.5-turbo-instruct
Answer relevancy	0.382168	0.28539	0.179251
Faithfulness	0.662306	0.775636	0.802007
Context precision	0.018633	0.174676	0.10133
Context recall	0.72443	0.809915	0.857978
Context relevancy	0.04149	0.082044	0.048747

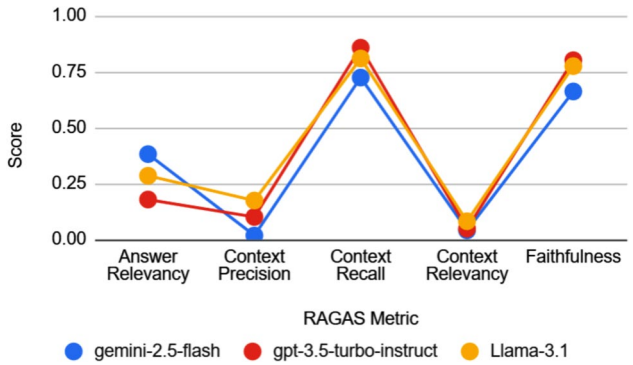


Fig. 7 Performance comparison of different LLMs by RAGAs metrics

or provides dangerously incomplete advice. Integrating such checks into the evaluation pipeline is a crucial step toward building a safer AI system.

Table 3 presents the performance of three language models across five evaluation criteria of RAG assessment. Figure 7 visually compares these results.

Llama 3.1 is the top overall performer, achieving the highest average score of 0.425532. It excels in context precision (0.174676) and context relevancy (0.082044), demonstrating a strong ability to provide accurate and relevant contextual information. GPT-3.5-turbo-instruct stands out for its high faithfulness (0.802007) and context recall (0.857978). This indicates that its responses are highly consistent with the provided context and that it is effective at retrieving all necessary information. Gemini-2.5-flash shows strength in answer relevancy, with the highest score of 0.382168, meaning its answers are the most pertinent to the questions. However, it lags in other areas, having the lowest average score (0.365805) among the models.

We consider choosing GPT-3.5 for our legal question-answering system primarily because of its exceptional performance in three key areas: MPNet score, faithfulness, and context recall. A good MPNet score suggests that GPT-3.5 is good at understanding the internal meaning, even if the words used are different. A high faithfulness score denotes that its responses are highly consistent with the provided context. A high context recall score ensures that the generated answer is comprehensive and does not miss any crucial details that were present in the source material. Models such as Gemini-2.5-flash and Llama3.1 also demonstrated themselves as promising alternatives, excelling in syntactic matching and answer relevancy, though requiring improvements in semantic understanding and faithfulness.

5 Conclusion and future work

In India, where legal literacy is often low and the judicial system is highly complex, a legal assistant powered by large language models (LLMs) can play a pivotal role in promoting a more equitable society. Navigating legal texts like the Bhartiya Nyaya Sanhita (BNS) or the Indian Penal Code (IPC) is often overwhelming and time-intensive, especially for non-lawyers facing legal challenges. LegalEase addresses this gap by offering a user-friendly platform that leverages advanced AI to make legal information more accessible. The system integrates foundational legal texts such as the BNS and IPC to provide accurate and authoritative answers. The latest official BNS document, including its amendments, has been used. The prompt was configured to prioritize BNS over IPC. However, this approach cannot handle future amendments, which is a limitation that can be addressed in future work. LegalEase efficiently maps user queries to relevant legal provisions using LLM-generated embeddings. This is done through an advanced information retrieval mechanism that prioritizes the most pertinent details. Custom-designed prompt templates further refine the LLM's ability to generate precise, context-aware responses. To ensure inclusivity, the platform supports multiple Indian languages, enhancing accessibility for a diverse audience. An intuitive Streamlit web app simplifies interaction by offering language options, sample queries, and a seamless user experience. This study evaluated three advanced large language models for legal Q&A systems and offers helpful guidance on selecting an effective model for such applications. While LegalEase shows great promise, there is significant room for improvement in its performance and design. Future development for the system should focus on enabling voice-based queries, expanding its legal database, and adopting more sophisticated retrieval techniques. Creating a user-friendly Android app will also extend accessibility to a broader audience. Continuous feedback from legal experts and practitioners will be vital to refining the system and ensuring the delivery of high-quality responses. By addressing these areas, LegalEase can evolve into a more robust tool, democratizing access to legal knowledge and empowering users across India.

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Acknowledgements This paper is a part of B.Tech Project work of Faiz Mubeen (APAAR id: 17915188345) under the guidance of Dr. Md. Asraful Haque, supported by IndiaAI Fellowship, Ministry of Electronics and Information Technology (MeitY), Govt. of India.

Author contribution FM conceptualized the study and designed the methodology. FM and AM performed the experiments and collected data. MN and NSU analyzed the results. MAH supervised the work and wrote the original draft. All authors read and approved the final manuscript.

Data availability The dataset used in the study can be provided on request.

Declarations

Ethical approval Ethical approval was not required for this study as it did not involve human or animal participants, or the collection of personally identifiable data.

Consent for publication This manuscript does not contain any individual person's data in any form (e.g., details, images, or videos); therefore, consent for publication is not applicable.

Conflict of interest The authors declare no competing interests.

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